

Turkey DNA extraction

Protocol for use of Zymo Research [Quick-DNA Miniprep plus kit](#) for extracting DNA from Nucleated Turkey Blood samples.

Materials

- Zymo Research Quick-DNA Miniprep plus kit
- Whole Blood sample preserved in RNA/DNA shield
- 1000uL pipette w/ tips
- 200uL pipette w/ tips
- Eppendorf tubes

For Lysed Samples in DNA RNA shield

- ☐ Thaw sample entirely and vortex briefly to make sure sample is homogenous.
- ☐ Take 400uL of sample and transfer to a DNase-free 1.7mL tube. Add 20uL of Proteinase K to this sample.
- ☐ Vortex for 10-15 seconds to mix thoroughly and incubate at room temperature for 20 minutes or overnight.
 - Note: Need to test overnight digestion. Normal processing protocol allows for overnight digestion at 55C.
- ☐ Add 1 volume (420uL for maximum input volume) of Genomic Binding Buffer to the digested sample. Vortex for 10-15 seconds.
- ☐ Transfer mixture to a Zymo-Spin IIC-XLR Column in a collection tube. Centrifuge at $\geq 12,000\times g$ for 1 minute or until completely cleared. Discard the flow through and collection tube.
- ☐ Add 400uL **DNA Pre-Wash Buffer** to the spin column in a **new** collection tube.
- ☐ Add 700uL **g-DNA Wash Buffer** to the spin column. Centrifuge at $\geq 12,000\times g$ for 1 minute. Empty the collection tube.
- ☐ Add 200uL **g-DNA Wash Buffer** to the spin column. Centrifuge at $\geq 12,000\times g$ for 1 minute. Discard the collection tube and flow through.

- ☐ Transfer the spin column to a clean microcentrifuge tube. Add 50uL of DNA Elution Buffer or Nuclease-free water directly to the matrix. Incubate for 5 minutes at room temperature, then centrifuge at maximum speed for 1 minute to elute the DNA. The eluted DNA can be used immediately for molecular based applications or stored at $\leq -20^{\circ}\text{C}$ for future use

For Nucleated Blood Pellets

- ☐ Thaw sample completely and centrifuge at 5,000xg for 30 seconds to pellet blood cells.
- ☐ While sample is thawing, prepare the following in a separate 1.7mL microcentrifuge tube:

BioFluid & Cell Buffer (Red)	200uL
ProteinaseK	20uL
DNA Elution Buffer	200uL

- ☐ Placing the tip of a pipette at the bottom of the sample tube, **slowly** draw up 10uL of blood from the pellet at the bottom of the tube. Do your best not to draw up any DNA/RNA Shield. **Slowly** dispense the blood into the tube you prepared previously.
- ☐ Mix tube thoroughly then incubate at 55°C for 20 minutes or overnight.
- ☐ Add 1 volume of Genomic Binding Buffer to the tube and mix thoroughly by pipetting up and down **and** by vortexing. Ensure the sample is homogenous before continuing.
- ☐ Transfer the mixture to a Zymo-Spin IIC-XLR Column in a Collection tube. Centrifuge at $\geq 12,000\times g$ for 1 minute or until the whole sample has passed through. Discard the Collection Tube with the flow through.
- ☐ Add 400uL **DNA Pre-Wash** to the spin column in a new collection tube. Centrifuge at $\geq 12,000\times g$ for 1 minute. Empty the collection tube.
- ☐ Add 700uL **g-DNA Wash Buffer** to the spin column. Centrifuge at $\geq 12,000\times g$ for 1 minute. Empty the Collection tube.
- ☐ Add 200uL **g-DNA Wash Buffer** to the spin column directly on the matrix. Centrifuge at $\geq 12,000\times g$ for 1 minute. Discard the collection tube and flow through.
- ☐ Transfer the spin column to a fresh DNase-free 1.7uL microfuge tube. Add 50uL **DNA Elution Buffer** or Nuclease-Free water directly on the matrix. Incubate for 5 minutes at room temperature, then centrifuge at top speed for 1 minute to elute the DNA. The eluted DNA can be used immediately for molecular based applications or stored at $\leq -20^{\circ}\text{C}$ for future use.

Protocol Notes

- If using a sample input of <50uL for the lysed sample protocol, increase volume to 50uL using DNA Elution Buffer or an isotonic buffer like PBS before continuing.